

The effectiveness of cryotherapy and cooled topical anesthesia compared with conventional topical anesthesia in alleviating intraoral injection pain: A systematic review and meta-analysis

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Abstract

Background: Injection pain remains a significant concern in dental procedures, often leading to patient anxiety and reluctance to seek necessary care.

Aim: This study aimed to assess the effectiveness of cryotherapy in reducing injection pain compared with topical anesthesia during intraoral local anesthesia.

Design: MEDLINE, Web of Science, EMBASE, Scopus, and Cochrane's CENTRAL databases were searched up to November 2023. Inclusion criteria involved randomized clinical trials aligned with the PICO question. Study selection, data extraction, and risk of bias assessment using the ROB-2 tool were performed. The results were synthesized through a random-effects inverse variance meta-analysis. The primary outcomes assessed were the Visual Analogue Scale (VAS) and Sound Eye Motor (SEM). Subgroup analysis was conducted for children and adults.

Results: In the meta-analysis, 31 RCTs involving 2470 subjects were included. Twenty-seven studies demonstrated cryotherapy's significant superiority over topical anesthesia in reducing injection pain via VAS in adults ($p = .01$), children ($p = .01$), and combined age groups ($p < .001$). Additionally, cryotherapy significantly outperformed topical anesthesia in reducing pain via SEM in children ($p = .04$) and combined age groups ($p = .03$) across 13 studies, with no significant difference in adults ($p = .51$). Furthermore, cooled topical anesthesia also outperformed room temperature topical anesthesia ($p < .001$). The certainty of the results, however, is of very low quality.

Conclusions: Our results indicate that cryotherapy significantly reduces injection pain compared with topical anesthesia, especially in children and combined age groups, but is less effective in adults. Additionally, cooled topical anesthesia is more effective than room temperature topical anesthesia. Thus, cryotherapy is a and potentially superior alternative to topical anesthesia, particularly for children.

KEYWORDS

behavior control, child, cryotherapy, dental anesthesia, ice, pain

1 | INTRODUCTION

Pain is a complex experience that involves both physiological and psychological responses to noxious stimuli, leading individuals to protect themselves by avoiding harm.¹ The psychological component is particularly relevant in dental settings, where fear and anxiety can greatly influence the perception of pain.

Dental fear is a prevalent issue worldwide, often resulting in the avoidance of dental care.² One of the primary contributors to this dental anxiety is the painful injection experience and inadequate pain control.³ This fear affects both children and adults, but it is particularly pronounced in children. It significantly prevents them from seeking dental care, resulting in long-term issues that affect their quality of life.⁴

Local anesthesia plays a crucial role in dental procedures to ensure a painless experience for patients and alleviate their anxiety. Although complete painless injection of local anesthesia is almost impossible to achieve, several methods can reduce the pain associated with it.⁵ These methods include the use of topical anesthesia (eg, Lidocaine), warming local anesthetic agents, buffering the local anesthetics, adjusting the injection rate, applying counter-irritation, using distraction techniques, and applying pressure to the injection site.⁵

Another effective approach to reduce injection pain during dental procedures is cryotherapy. Cryotherapy involves precooling the injection site, resulting in vasoconstriction, inhibition of nerve impulses, activation of pain-inhibitory pathways, reduction in edema, and slower release of inflammatory mediators.^{6,7} Moreover, the application of cryotherapy not only offers physiological benefits but also serves as a distraction for patients, diverting their attention from discomfort.⁸

Various modalities can be used to administer cryotherapy, each with its own advantages and disadvantages. One common method is the application of an ice pack directly to the injection site. This technique is easy to use, widely available, and cost-effective, providing adequate cooling and numbing effects. It, however, requires a few minutes to be effective⁹ and can cause discomfort if applied for too long, as well as being messy as the ice melts. Another approach is the use of a refrigerant spray on a cotton applicator. This method offers rapid onset of cooling and easy control of the application area, without the mess associated with melting ice. Nevertheless, the cooling effect is shorter in duration, and there is a risk of tissue irritation if not used correctly.¹⁰ Gel packs, which can be cooled in a freezer and then applied to the injection site, offer the advantage of being reusable and conforming to the shape of the injection area, providing consistent cooling without melting.

Why this paper is important to paediatric dentists

- Paediatric dentists will find the research valuable as cryotherapy emerges as a safe and accessible option, providing an additional tool in their arsenal for enhancing patient care and overall satisfaction in paediatric dental practice.
- Implementing cryotherapy in paediatric dentistry could contribute to creating a more positive and comfortable dental experience for children, potentially reducing anxiety and fear associated with injections and promoting better compliance with dental procedures.
- Future research should focus on different injection sites and techniques, standardizing application times, and examining various demographic groups to provide a comprehensive understanding of cryotherapy's implications to enhance evidence-based practices for pain management in paediatric dental settings.

They, however, may take longer to achieve the desired cooling effect compared with refrigerant sprays.

Despite the potential benefits of cryotherapy in alleviating injection pain, there is an ongoing debate regarding its actual effectiveness.^{11,12} This debate, however, remains ongoing, with conflicting findings from various studies contributing to the uncertainty surrounding its efficacy. Therefore, to address this discrepancy and provide clarity on the matter, a systematic review and meta-analysis is necessary. The primary objective of this review was to assess the impact of cryotherapy on injection pain perception by using the Visual Analogue Scale (VAS) and Sound Eye Motor (SEM). By synthesizing existing evidence, this review aimed to enhance our understanding of this technique and assist clinicians in selecting the most suitable approach.

2 | MATERIALS AND METHODS

2.1 | Study design and study protocol

In conducting this systematic review and meta-analysis, the guidelines outlined in the Cochrane Handbook for Systematic Reviews of Interventions were adhered to, and followed the principles established by the PRISMA model.¹³ Furthermore, the research protocol was proactively registered with the International Prospective Register of Systematic Reviews, under the registration number:

PROSPERO (#CRD42024495929). The primary objective of this review was to evaluate the effectiveness of cryotherapy in reducing injection pain compared with topical anesthesia in patients undergoing intraoral local anesthesia for dental procedures. Additionally, a secondary objective was to compare the effectiveness of cooled topical anesthesia against room temperature topical anesthesia.

2.2 | Eligibility criteria

Our study selection process followed the PICO approach, which involved identifying patients undergoing dental procedures (Population), assessing the impact of cryotherapy (Intervention) compared with alternative pain management methods like topical anesthesia (Comparison), and prioritizing outcomes related to injection pain perception (Outcome). All pertinent clinical trials and cohort studies that aligned with the PICO model were included, without restrictions based on publication date or language. Studies not involving human subjects were excluded from consideration.

2.3 | Search strategy and data sources

Our comprehensive literature search involved multiple databases, including MEDLINE, Web of Science, EMBASE, Scopus, and Cochrane's CENTRAL. To ensure thoroughness, supplementary searches were conducted in Google Scholar and ResearchGate to capture any additional relevant studies. Articles published from their inception up to November 2023 were searched. Additionally, manual searches were conducted in prominent dentistry journals, reviewed relevant reference lists, and explored online trial directories. When study registrations provided leads to relevant information, correspondence was engaged with the original authors. The keywords used in this extensive search process included "cryotherapy," "cold therapy," and "injection." Detailed information about the search methodology for each database is available in [Table 1](#).

2.4 | Selection process for studies

Adhering to the PICO principle, two independent reviewers (MG and PS) conducted a thorough review of the study titles and abstracts. Initially, a pilot selection process, involving initial training sessions and regular meetings, was conducted to ensure alignment and calibration between the reviewers. In instances of disagreement, discussions were held, and if a consensus could not be reached, a

third expert (ASS) was consulted for resolution to ensure a standardized approach for study selection. Studies that did not align with the PICO criteria were excluded. Following this initial screening, the full texts of the chosen studies were obtained, and their eligibility was assessed in accordance with the PICO question.

2.5 | Data extraction

Data extraction from the eligible studies was performed using a pre-piloted form. Two authors (PS and MG) independently conducted the data extraction, and any discrepancies were resolved through collaborative discussion. The extracted data included details such as the publication year, study design, participant demographics, the type and site of anesthesia, the type of anesthetic agent, the type of cryotherapy, cryotherapy application duration, needle gauge, and the outcomes assessed, which encompassed measures such as the VAS and SEM. In cases of incomplete data, efforts were made to contact the original authors for clarification.

2.6 | Risk of bias and quality assessment

To objectively assess potential biases within the included studies, the Cochrane Risk of Bias 2.0 tool was utilized. This assessment was conducted independently by two reviewers (PS and ASS), by examining five key criteria: randomization process, deviations from intended interventions, missing outcome data, outcome measurement, and selection of reported results.¹⁴ If any of the five criteria were identified as having a high risk or some concerns, the study would be categorized as having a high risk of bias or some concerns, respectively. Consequently, any studies with an overall high risk of bias were excluded from the meta-analysis.

Additionally, the quality and reliability of the presented evidence were evaluated using the Grading of Recommendations Assessment, Development, and Evaluation (GRADE) system. This system categorizes evidence into four levels: high, moderate, low, or very low. Various factors contribute to the evaluation of certainty and quality, including study design, risk of bias, inconsistency in results, indirectness of evidence, imprecision of results, and other elements such as publication bias.¹⁵

2.7 | Data integration and analysis

The primary outcomes subjected to meta-analysis encompassed the VAS and SEM. A comparative analysis

TABLE 1 Databases, applied search strategy, and numbers of retrieved studies.

Database of published trials, dissertations, and conference proceedings	Search strategy used	Hits
MEDLINE searched via PubMed searched on November 1, 2023, via www.ncbi.nlm.nih.gov/sites	((“Cryotherapy”[Mesh]) OR Cryotherapy* OR (Cold Therap*) OR (“Ice”[Mesh]) OR ice OR cool* OR precool*) AND (Injection* OR Anesthesia OR (“Anesthesia”[Mesh]) OR (“Anesthetics”[Mesh]) OR (“Anesthesiology”[Mesh]) OR anesthetic OR anesthe* OR infiltration OR (nerve block) OR (“nerve block”[MeSH Terms])) AND (oral OR mouth OR teeth OR tooth OR dent* OR maxilla* OR mandib* OR molar* OR incis* OR palatal* OR alveolar)	1304
Web of science Core Collection was searched via web of knowledge on November 2, 2023, via apps.webofknowledge.com	(cryotherapy* OR (cold AND therap*) OR ice OR cool* OR precool*) AND (injection* OR anesthesia OR anesthetic OR anesthe* OR infiltration OR (nerve AND block)) AND (oral OR mouth OR teeth OR tooth OR dent* OR maxilla* OR mandib* OR molar* OR incis* OR palatal* OR alveolar)	614
EMBASE searched via Ovid on November 2, 2023, via http://ovidsp.dc2.ovid.com	(cryotherapy* OR ((‘cold’/exp. OR cold) AND therap*) OR ‘ice’/exp. OR ice OR cool* OR precool*) AND (injection* OR ‘anesthesia’/exp. OR anesthesia OR ‘anesthetic’/exp. OR anesthetic OR anesthe* OR ‘infiltration’/exp. OR infiltration OR ((‘nerve’/exp. OR nerve) AND block)) AND (oral OR ‘mouth’/exp. OR mouth OR ‘teeth’/exp. OR teeth OR ‘tooth’/exp. OR tooth OR dent* OR maxilla* OR mandib* OR molar* OR incis* OR palatal* OR alveolar) AND (cryotherapy*:ti,ab,kw OR (cold:ti,ab,kw AND therap*:ti,ab,kw) OR ice:ti,ab,kw OR cool*:ti,ab,kw OR precool*:ti,ab,kw) AND (injection*:ti,ab,kw OR anesthesia:ti,ab,kw OR anesthetic:ti,ab,kw OR anesthe*:ti,ab,kw OR infiltration:ti,ab,kw OR (nerve:ti,ab,kw AND block:ti,ab,kw)) AND (oral:ti,ab,kw OR mouth:ti,ab,kw OR teeth:ti,ab,kw OR tooth:ti,ab,kw OR dent*:ti,ab,kw OR maxilla*:ti,ab,kw OR mandib*:ti,ab,kw OR molar*:ti,ab,kw OR incis*:ti,ab,kw OR palatal*:ti,ab,kw OR alveolar:ti,ab,kw)	948
Scopus searched via Scopus on November 1, 2023, via https://www.scopus.com	TITLE-ABS-KEY ((cryotherapy* OR (cold AND therap*) OR ice OR cool* OR precool*) AND (injection* OR anesthesia OR anesthetic OR anesthe* OR infiltration OR (nerve AND block)) AND (oral OR mouth OR teeth OR tooth OR dent* OR maxilla* OR mandib* OR molar* OR incis* OR palatal* OR alveolar))	1492
Cochrane Central Register of Controlled Trials searched via the Cochrane Library Searched on November 3, 2023, via www.thecochranelibrary.com	(cryotherapy* OR (cold AND therap*) OR ice OR cool* OR precool*) AND (injection* OR anesthesia OR anesthetic OR anesthe* OR infiltration OR (nerve AND block)) AND (oral OR mouth OR teeth OR tooth OR dent* OR maxilla* OR mandib* OR molar* OR incis* OR palatal* OR alveolar)	566
Total		4924

of cryotherapy method and topical anesthesia group was conducted. Additionally, a secondary comparison between cooled topical anesthesia and room temperature topical anesthesia was conducted. Furthermore, a subgroup analysis was conducted for two age groups: children (patients under 18 years of age) and adults (patients aged 18 and above), when applicable. To account for the continuous nature of the outcomes, weighted pooled standard mean differences (Hedges's g effect size) were calculated using inverse variance meta-analysis with random-effect models. Additionally, for the studies that employed a control group without any intervention as a comparison instead of topical anesthesia, a sensitivity analysis was conducted. To assess study heterogeneity, the chi-squared Q -test and the I^2 metric were utilized. All meta-analyses were conducted using the Stata software (Version 17; StataCorp LLC, TX, USA).

3 | RESULTS

3.1 | Study selection

After a thorough search across online databases, gray literature, and relevant journals, a total of 4924 studies were initially identified. Following the removal of duplicate entries, 2806 studies remained, and their titles and abstracts underwent meticulous evaluation. Applying the inclusion and exclusion criteria alongside the PICO model, 2762 articles were subsequently excluded. A thorough assessment of the full texts of the remaining 44 studies, along with an additional six studies identified through manual searches, was conducted for eligibility. This process led to the identification of 37 studies that met the specified criteria. Consequently, our qualitative assessments were based on 37 studies,^{6,9–12,16–47} and data from 31 of these studies were incorporated into our meta-analysis.

lyses.^{6,10–12,16,17,19–23,25–27,29–38,40–45,47} A detailed flow chart illustrating our study selection process is presented in Figure 1.

3.2 | Study characteristics

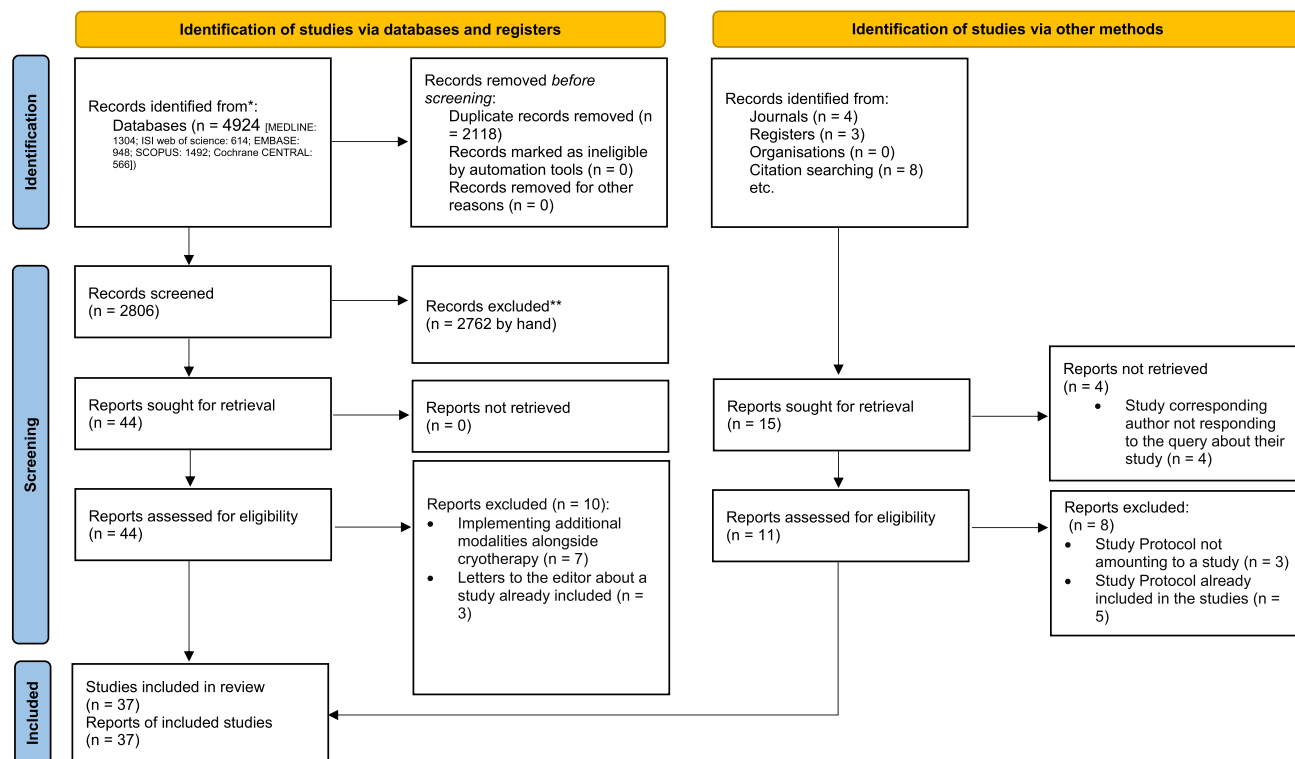
Among the 37 selected studies with randomized clinical trial (RCT) designs, 19 were split-mouth trials. The sample size of these studies varied from 9 to 200 subjects, totaling 2368 patients. Notably, among 31 studies included in our analysis, 23 studies involved subjects under 18 years, categorized as children in our quantitative analysis, whereas 14 studies involved adults aged 18 and above, categorized as adults in our analysis. Four studies employed two sets of groups with distinct participants.^{23,31,38,40} Specifically, one study compared infiltration versus nerve block, exploring two modalities.²³ Two studies investigated buccal infiltration versus palatal infiltration, each involving two modalities.^{31,38} The last study examined two different application times.⁴⁰ Both sets of groups from these studies were included in our quantitative analysis.

Various types of cryotherapy were employed across the studies, including refrigerant spray on a cotton ball, ice packs, frozen capsules, and ice cubes. Three studies utilized cooled topical anesthesia,^{41,43,47} with two of them employing 20% benzocaine gel^{41,43} and one using 5% lidocaine gel.⁴⁷ The application time ranged from 5 s to 4 min, although one study did not report the application time.¹¹ The site of injection varied, encompassing nerve blocks or infiltrations in either the upper or lower jaw, and needle gauges range from 24 to 31.

Regarding comparison groups, diverse comparisons were implemented. Most studies used topical anesthesia with different compositions and percentages, whereas some employed a control group with no intervention for reducing injection pain.^{10,16,17,23,44} A few studies used other comparisons, such as honey,¹⁷ counter-irritation,⁴⁶ vibration devices,^{19,38,42} and sucrose solution,²⁴ though their limited usage precluded their inclusion in our quantitative analysis.

The assessment methods also varied, with the VAS and SEM being the most commonly employed. Some studies used different subjective scales, categorized under VAS

PRISMA 2020 flow diagram for new systematic reviews which included searches of databases, registers and other sources



*Consider, if feasible to do so, reporting the number of records identified from each database or register searched (rather than the total number across all databases/registers).

**If automation tools were used, indicate how many records were excluded by a human and how many were excluded by automation tools. From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ* 2021;372:n71. doi: 10.1136/bmj.n71. For more information, visit: <http://www.prisma-statement.org/>

FIGURE 1 PRISMA study flow diagram. *Consider, if feasible to do so, reporting the number of records identified from each database or register searched (rather than the total number across all databases/registers). **If automation tools were used, indicate how many records were excluded by a human and how many were excluded by automation tools. From: Page et al.⁶⁶ For more information, visit: <http://www.prisma-statement.org/>.

assessment, such as Heft Parker-Visual Analogue Scale, Wong Baker's Faces Pain Rating Scale, Faces Pain Scale, Numerical Rating Scale, and Colored Analog Scale.

Regarding the washout period in split-mouth studies, five studies considered a 1-week or at least 1-week period,^{12,25,30,45,47} whereas another five studies employed a 2-week or at least 2-week period.^{10,31,36,40,43} Three studies conducted both interventions within the same appointment, randomly dividing participants into two groups, with each group receiving one of the interventions first, thus pooling the results to prevent any reporting bias.^{26,33,34} In the studies involving children, six specifically included those attending their first dental appointment.^{12,18,29,36,41,46} The other studies, however, only reported the inclusion of children with a history of cooperation and no prior traumatizing experiences. Detailed characteristics and main results of the included studies are provided in Tables 2 and 3.

3.3 | Risk of bias within studies

Figure 2 illustrates the risk of bias in 37 studies. Six studies exhibited a high risk of bias and were consequently excluded from quantitative assessments. Among these, four were excluded due to missing outcome data and the lack of reported quantitative data, one due to serious bias in randomization and the absence of any mention of randomization, and one due to both missing outcome data and biased randomization. The remaining 31 studies demonstrated a moderate risk of bias.

Concerns related to randomization bias were raised in 13 studies, as the randomization method and allocation sequence concealment were not reported. In contrast, the remaining 18 studies provided information on these processes. The blinding of participants and personnel (performance bias) was not feasible due to the nature of the intervention, and its absence was assumed to have no impact on the outcomes. Given that the treatment was implemented in a single session, deviations from intended interventions were considered negligible in all studies. Except for the aforementioned five studies excluded due to missing outcome data, the remaining studies exhibited a low risk of bias and showed no attrition bias.

The blinding of outcome evaluators (detection bias) was not achievable since the treatment administrators and outcome evaluators were the same person in all studies. Due to the subjective nature of the outcomes, a moderate risk of measurement bias was considered in all studies. Regarding reporting bias and selective reporting, nine studies had registered study procedures, and all pre-specified primary and secondary outcomes were reported as planned. Consequently, these studies had a low risk of

bias, whereas the remaining studies encountered some issues due to unavailable protocols.

3.4 | Synthesis of results

3.4.1 | Cryotherapy versus topical anesthesia: Visual Analogue Scale meta-analysis

In a comprehensive meta-analysis encompassing 27 studies (with four of them including two separate sets of groups), involving a total of 2254 subjects, cryotherapy's effectiveness in reducing injection pain via the VAS was evaluated. Of these, 1127 received cryotherapy, whereas 1127 received topical anesthesia prior to injection.

Subgroup analyses based on interventions and age groups revealed that cryotherapy exhibited significant superiority over topical anesthesia in reducing injection pain in adult patients (SMD = -1.06, 95% CI = -1.81, -0.30; $p = .01$). Similarly, in the analysis involving children, cryotherapy demonstrated significantly better pain reduction compared with topical anesthesia (SMD = -0.71, 95% CI = -1.29, -0.14; $p = .01$). Overall, the meta-analysis revealed a statistically significant difference in VAS scores between the groups, with cryotherapy proving more effective than topical anesthesia (SMD = -0.87, 95% CI = -1.33, -0.41; $p < .001$). Moreover, no significant differences were reported between adult and children scores ($p = .48$). It is, however, crucial to note that the results' heterogeneity was statistically significant ($I^2 > 70\%$) for all the meta-analyses (Figure 3).

Despite the sustained significant heterogeneity ($I^2 = 96\%$), a sensitivity analysis revealed that cryotherapy continued to exhibit superior performance over the topical anesthesia group in reducing injection pain, even after the exclusion of studies by Abbasi et al., Wiswall et al., and Bose et al.,^{10,17,23} which employed a control group with no intervention as a comparison (SMD = -0.80, 95% CI = -1.30, -0.30; $p < .001$; Figure S1). The meta-analysis, with the omission of each individual study, consistently demonstrated significance, as illustrated in the case of excluding the three mentioned studies from the analysis (Figure S1).

3.4.2 | Cryotherapy versus topical anesthesia: Sound Eye Motor meta-analysis

Additionally, the effectiveness of cryotherapy on injection pain reduction via the SEM was evaluated in a meta-analysis involving 13 studies (with two of them including two separate sets of groups), comprising a total of 1306 patients. Of these, 653 received cryotherapy, whereas 653 received topical anesthesia prior to injection.

Analyses based on interventions and age groups indicated that cryotherapy did not show a significant difference compared with topical anesthesia in reducing injection pain in adult patients (SMD = -0.67 , 95% CI = -2.66 , 1.33 ; $p = .51$). Conversely, in the analysis involving children, cryotherapy exhibited significantly better pain reduction compared with topical anesthesia (SMD = -0.68 , 95% CI = -1.32 , -0.04 ; $p = .04$). Overall, the meta-analysis demonstrated a statistically significant difference in SEM scores between the groups, with cryotherapy being more effective than topical anesthesia (SMD = -0.68 , 95% CI = -1.29 , -0.07 ; $p = .03$). Furthermore, no significant differences were reported between adult and children scores ($p = .99$). It is, however, essential to acknowledge that the results' heterogeneity was statistically significant ($I^2 > 70\%$) for all the meta-analyses (Figure 4).

Following sensitivity analysis, the exclusion of the Bose et al. study,²³ which employed a control group with no intervention as a comparison, did not significantly alter the results. Despite the persistent high heterogeneity ($I^2 = 96\%$), cryotherapy continued to demonstrate superior pain reduction compared with topical anesthesia (SMD = -0.74 , 95% CI = -1.44 , -0.03 ; $p = .04$; Figure S2). The meta-analysis, with the omission of each individual study, consistently maintained significance in case of exclusion of the Bose et al. study from the analysis (Figure S2).

3.4.3 | Cooled topical anesthesia versus room temperature topical anesthesia: Visual Analogue Scale meta-analysis

Lastly, the effectiveness of cooled topical anesthesia on injection pain reduction via the VAS was evaluated in a meta-analysis involving three studies, comprising 152 patients. Of these, 76 received cooled topical anesthesia, whereas 76 received room temperature topical anesthesia prior to injection.

Analyses based on interventions indicated that cooled topical anesthesia exhibited a significant superiority compared with room temperature topical anesthesia in reducing injection pain in terms of VAS (SMD = -2.92 , 95% CI = -4.72 , -1.11 ; $p < .001$). It is, however, crucial to note that the heterogeneity of the results was statistically significant ($I^2 = 93\%$) for this meta-analysis, although a subgroup analysis was not conducted due to the limited number of studies (Figure 5).

Publication bias assessment

After visually inspecting the funnel plot and conducting subsequent analyses using Begg and Egger methods ($p > .1$), there was no discernible evidence of publication bias, as illustrated in Figure S3.

3.5 | Risk of bias across studies

Using the GRADE approach to assess the quality of evidence and the certainty of outcomes, it was determined that the evidence supporting the effectiveness of cryotherapy in both adults and children, across both VAS and SEM outcomes, was of very low quality. This low level of certainty was primarily due to a serious risk of bias related to the subjective nature of the outcome measurements, as well as significant inconsistencies in the findings. A detailed overview of the GRADE ratings is provided in Table S1.

4 | DISCUSSION

4.1 | Summary of evidence

Our study conducted meta-analyses to assess the effectiveness of cryotherapy in reducing injection pain, utilizing both the VAS and SEM. Subgroup analyses were performed based on interventions and age groups.

For VAS, our analysis revealed that cryotherapy demonstrated significant superiority over topical anesthesia in reducing injection pain in both adults and children. Nevertheless, taking into account potential limitations in the context of children's responses to the VAS is important. Children's given scores are more likely to be influenced by fear or previous painful experiences, and they often exhibit a tendency to choose extreme values (0 or 10).⁴⁸ This could be attributed to the limited experience children have with rating scales, possibly leading them to lean toward extreme values and struggle with the concept of a middle ground, potentially introducing less reliability compared with adults.

Concerning SEM, the analysis involving children indicated that cryotherapy exhibited significantly better pain reduction compared with topical anesthesia. In adults, however, cryotherapy did not show a significant difference in SEM. This could be attributed to SEM being more suitable for assessing pain perception in children, as adults typically exert more control over their reactions and sounds in response to pain. Additionally, the limited number of studies in this subgroup analysis should be considered when interpreting the results.

Following sensitivity analysis, the exclusion of studies using a control group with no intervention as a comparison did not significantly alter the results for both VAS and SEM, and the overall impact of cryotherapy remained significantly better than topical anesthesia.

Overall, cryotherapy emerges as a viable and accessible alternative, even outperforming topical anesthesia, and appears to be a safe option for clinical practice. Moreover,

TABLE 2 Characteristics of the included studies involving children.

Study ID	Study design	Sample size, no.	Gender, no.	Mean (SD) age, years	Cryotherapy type	Application time	Type/Site of anesthesia	Needle gauge	Comparison	Assessment type	Wash out period	First dental visit	Main results
Aminabadi, 2009	RCT	160	F 88 M 72	5.25 (NR)	Ice tube	2 min	Inferior alveolar nerve block/Mandible	27	Topical Anesthesia (benzocaine)	SEM	NA	Yes	Cryotherapy > Topical anesthesia
Ghaderi, 2013	Split-mouth	50	F 26 M 24	8.94 (0.76)	Refrigerated cartridges which were filled with water	1 min	Buccal infiltration/Maxillary canine	27	Topical anesthesia (20% benzocaine)	VAS, SEM	Same appointment	NR	Cryotherapy > Topical anesthesia
Lathwal, 2015	RCT split-mouth	160	NR	5–8	Ice cone, Refrigerant spray	1 min, 5 s	Inferior alveolar nerve block or greater palatine nerve block/Mandible or maxilla	25	Topical anesthesia (benzocaine)	VAS, SEM	NR	NR	Cryotherapy > Topical anesthesia
Mohiuddin, 2015	Split-mouth	100	F 44 M 56	8–12	Ice cubes	1 min	Primary maxillary anterior teeth	NR	Topical anesthesia (2% lidocaine)	VAS	NR	NR	Cryotherapy > Topical anesthesia
Aminah, 2017	RCT	40	NR	7–13	Ice cubes	1 min	NR	NR	Topical anesthesia (20% Benzocaine), vibration, buffered anesthetic agent	WB-FPRS	NA	NR	Cryotherapy > Topical anesthesia
Hameed, 2018	Split-mouth	50	F 28 M 22	8.94 (0.76)	Refrigerant spray (tetrafluoroethene)	10 s	Inferior alveolar nerve block/Mandible	26	Topical anesthesia (5% lidocaine)	VAS, SEM	NR	NR	Cryotherapy > Topical anesthesia
Vafaei, 2019	RCT	99	NR	6–10	Ice, Refrigerant spray	1 min, 5 s	Buccal infiltration/Primary maxillary molar	25	Counter-irritation, topical anesthesia (20% benzocaine)	VAS, SEM	NA	Yes	Topical anesthesia > Cryotherapy, counter-irritation
Padmanabhan, 2019	RCT	57	NR	6–13	Ice	1 min	Inferior alveolar nerve block/Mandible	NR	Topical anesthesia, vibration	WB-FPRS	NA	NR	Vibration > Cryotherapy > Topical anesthesia
Bose, 2019	RCT split-mouth	100	NR	6–14	Ice tube	1 min	NR	NR	Control	VAS, WB-FPRS, SEM	NR	NR	Cryotherapy > Control
Soni, 2020	Split-mouth	50	NR	7–12	Ice	4 min	Buccal and palatal infiltration/Maxillary primary molars	NR	Topical anesthesia (lidocaine)	VAS, SEM	At least 1 week	NR	Cryotherapy > Topical anesthesia

TABLE 2 (Continued)

Study ID	Study design	Sample size, no.	Gender, (SD) age, years	Mean	Cryotherapy type	Application time	Type/Site of anesthesia	Needle gauge	Comparison	Assessment type	Wash out period	First dental visit	Main results
Chilakamuri, 2020	Split-mouth	30	F 10 M 20	8.1 (NR)	The pencil of ice was prepared by cutting away the adaptor portion of the 2 mL syringe until the first millimeter marking on the barrel	2 min	Greater palatine block/Maxilla	27	Topical anesthesia (5% lidocaine)	FLACC, CAS, pulse oximeter, treatment satisfaction	At least 1 week	Yes	Cryotherapy > Topical anesthesia
Anantharaj, 2020	RCT	60	F 27 M 33	9.45	Ice cone	1 min	NR	NR	Topical anesthesia (benzocaine), clove-papaya extract	WB-FPRS, SEM	NA	NR	Cryotherapy = Topical anesthesia = Clove-papaya extract
Lakshmanan, 2021	Split-mouth	30	F 13 M 17	8.28 (0.1)	Ice pack	2 min	Buccal infiltration/Maxilla	27	Topical anesthesia (20% Banzocaine)	SEM, VAS	2 weeks	Yes	Cryotherapy > Topical anesthesia
Prathyusha, 2021	Split-mouth	16	NR	8–10	Cooled topical anesthetic gel (20% benzocaine)	1 min	Buccal and palatal anterior infiltration/Maxilla	24	Topical anesthesia (20% benzocaine)	WB-FPRS, pulse oximeter, pulse rating, FLACC	At least 2 weeks	NR	Cryotherapy > Topical anesthesia
Havale, 2021	RCT	56	F 22 M 34	9.2	Ice cones	1 min	Buccal infiltration/Maxilla	27	Topical anesthesia (2% lidocaine), clove oil gel, betel leaf gel	WB-FPRS, SEM	NA	Yes	Cryotherapy = Clove oil gel = Betel leaf gel > Topical anesthesia
Islam, 2021	RCT	64	F 27 M 37	9.68 (1.41)	Ice pack was made by filling the finger of a latex glove with water,	1 min	NR	27	Topical anesthesia (20% benzocaine)	SEM	NA	NR	Cryotherapy > Topical anesthesia
Jafarrehmani, 2021	Split-mouth	14	F 6 M 8	7–10	Frozen carpool	1 min	Bilateral palatal anesthesia/Maxilla	30	Topical anesthesia (20% benzocaine)	WB-FPRS, SEM	Same appointment	NR	Cryotherapy = Topical anesthesia

(Continues)

TABLE 2 (Continued)

Study ID	Study design	Sample size, no.	Gender, no.	Mean (SD) age, years	Cryotherapy type	Application time	Type/Site of anesthesia	Needle gauge	Comparison	Assessment type	Wash out period	First dental visit	Main results
Singh, 2022	RCT	60	NR	6–11	Ice cubes	1 min	NR	30	Control, control and VR, topical anesthesia (5% lidocaine), topical anesthesia (5% lidocaine) and VR, cryotherapy and VR	WB-FPRS	NA	NR	Topical anesthesia and VR > Cryotherapy and VR > Topical anesthesia > Control and VR > Control > Cryotherapy
Dhingra, 2022	Split-mouth	132	NR	7–11	Ice cube	2 min	Inferior alveolar nerve block/Mandible	30	30% sucrose solution, popsicle (cryotherapy + sucrose), topical anesthesia (2% lidocaine)	WB-FPRS, SEM, FLACC, vital parameters	NR	NR	Popsicle > Topical anesthesia
Virdikar, 2022	Split-mouth	30	NR	8	Cooled topical anesthetic gel (5% lidocaine)	1 min	NR	NR	Topical anesthesia (5% lidocaine)	VAS, pulse rate	1 week	NR	Cryotherapy > Topical anesthesia
Nat, 2023	Split-mouth	30	NR	6–10	Cooled topical anesthetic gel (5% lidocaine)	1 min	Buccal infiltration/Maxilla	27	Topical anesthesia (5% lidocaine)	VAS	NR	Yes	Cryotherapy > Topical anesthesia
Mittal, 2023	RCT	90	F 39 M 51	9.12 (2.42)	Ice prepared by filling a 2.0-mL plastic disposable syringe with tap water and freezing it in a freezer	1 min	Buccal and palatal infiltration/Maxilla	30	Topical anesthesia (5% lidocaine), vibration	FPS, SEM	NA	NR	Cryotherapy = Vibration = Topical anesthesia

Note: > superior effectiveness; = comparable effectiveness.

Abbreviations: CAS, Colored Analogue Scale; F, female; FLACC, face legs activity cry and consolability; FPS, Faces Pain Scale; M, male; NA, not applicable; NR, not reported; RCT, randomized clinical trial; SEM, Sound Eye Motor; VAS, Visual Analogue Scale; WB-FPRS, Wong Baker's Faces Pain Rating Scale.

the cold temperature may act as a distraction, enhancing its potential benefits.

Furthermore, an additional analysis comparing cooled topical anesthesia and room temperature topical anesthesia confirmed the superiority of cooled topical anesthesia via VAS, harnessing the benefits of both the anesthetic agent and the cold temperature. Clinicians may find this technique beneficial due to its availability, feasibility, and safety.

Upon reviewing the literature, a systematic review by Amrollahi et al.⁴⁹ included six studies. They concluded that cryotherapy is more effective than topical anesthesia agents, but the evidence supporting the use of cold agents is limited and of low quality, and emphasized the need for more studies to validate their findings.

4.2 | Factors affecting the outcome and other considerations

The reviewed studies employed different study designs, each with its own set of strengths and limitations. Randomized clinical trials were utilized in some studies, where two distinct groups were randomly assigned to either cryotherapy or topical anesthesia. A challenge, however, arises in comparing pain perception through subjective measures like the VAS or similar scales, which can exhibit variations among individuals. Even assessments like the SEM remain subjective to the observer, and comparing scores between different patient groups raises potential concerns.

On the contrary, other studies opted for a split-mouth design, involving patients requiring intraoral anesthetic injections on both sides. One side was randomly assigned to cryotherapy, whereas the other received topical anesthesia, each in different sessions. Carry-over effects occur when the impact of one intervention persists during the evaluation of another. To prevent this, study periods can be separated, allowing participants to be free from the influence of the previous intervention. This separation period is referred to as the washout period.⁵⁰ Whereas this design offers the advantage of obtaining two scores from the same group of individuals, it introduces its own limitation. If the washout period is not adequate and when two interventions are conducted in separate sessions, the pain experienced in the first session can influence the perception of pain in the subsequent session, regardless of the intervention.⁵¹ Moreover, in the context of individuals undergoing their initial dental procedures, fear can significantly impact pain perception. In the first session, patients may experience heightened pain due to anxiety, and subsequent sessions may witness desensitization.⁵¹ It is, however, noteworthy that this limitation is absent in RCT designs.

Gender differences in pain perception have been explored extensively in the literature. There is evidence suggesting that men and women may experience and express pain differently. Studies indicate that, on average, women may be more sensitive to pain than men when exposed to the same stimuli.⁵² Hormonal factors, particularly fluctuations in estrogen levels, have been proposed as contributors to gender differences in pain perception, for instance, women may perceive more pain during the luteal phase of the menstrual cycle.⁵² It is crucial to recognize that individual responses to pain can vary significantly, and these variations may be influenced by a range of factors, including psychological and biological aspects.⁵³ Additionally, societal expectations and cultural norms are key contributors to how individuals express and report pain. Nevertheless, interestingly, this gender difference in pain perception seems to be absent in children and adolescents but becomes more prominent in adults.⁵⁴

Notably, in our systematic review, the assessment of pain perception between genders was not included in any of the studies. Consequently, we were unable to conduct a subgroup analysis based on gender to determine whether there is a discernible gender difference in pain perception in the context of the intraoral injection pain.

Another influential factor potentially impacting outcomes is the application time, a crucial consideration in both topical anesthesia and cryotherapy. Existing evidence supports a standardized and optimal 2-min application time for topical anesthesia, recognized as effective across studies.⁵⁵ In contrast, cryotherapy lacks a universally established application duration, with the included studies displaying variations ranging from 5 s to 4 min.

The diversity in cryotherapy application times may be intricately linked to the specific type of cryotherapy employed in the studies. Different techniques were used in the studies, including direct ice contact, frozen carpules, freezing water-filled latex glove fingers, and using refrigerant spray.^{28,30,33,56} It is important to note that these methods involve materials with different thermal conductivity properties which could affect the application time. For instance, direct ice contact requires a shorter application time due to its higher thermal conductivity. On the contrary, frozen carpules, which involve glass with a lower thermal conductivity than ice, need a longer application time, and likewise, using a frozen latex glove requires a more extended application period.⁵⁷ Additionally, when considering the use of refrigerant spray, clinicians must be mindful of its potential adverse effects. Directly applying refrigerant spray on a cotton ball for a mere 10 s with pressure can lead to tissue damage,¹⁰ and therefore, a careful approach involving a shorter application time is advisable in such cases. Furthermore, Vafaei et al.⁴⁶ reported that

TABLE 3 Characteristics of the included studies involving adults.

Study ID	Study design	Sample size, no.	Gender, no.	Mean (SD) age, years	Cryotherapy type	Application time	Type/Site of anesthesia	Needle gauge	Comparison	Assessment type	Wash out period	Main results
Kosaraju, 2009	Split-mouth	16	F 5 M 11	42.8	Refrigerant 1,1,1,3,3-pentafluoropropane/1,1,1,2,2-tetrafluoroethane	5 s	Bilateral posterior palatal anesthesia/Maxilla	30	Topical anesthesia (20% benzocaine)	VAS	Same appointment	Cryotherapy > Topical anesthesia
Wiswall, 2014	RCT split-mouth	42	F 21 M 21	>18	Cotton tip applicator impregnated with a three-second sprayed application of endo-ice	10 s	Greater palatine block/Maxilla	27	Control, pressure with cotton tip applicator, pressure with cotton tip applicator + topical anesthesia (20% benzocaine)	HP-VAS	At least 2 weeks	Cryotherapy + Pressure = Control = Pressure = Pressure + Topical anesthesia
Bhadoria, 2017	Split-mouth	33	F 9 M 24	>18	Ice cube	3 min	Infiltration/Maxilla	NR	Topical anesthesia (lidocaine)	HP-VAS, SEM	NR	Cryotherapy > Topical anesthesia
Hindocha, 2019	RCT split-mouth	40	F 21 M 19	15.6 (2.2)	Ice	1 min	Buccal and palatal infiltration/Maxilla	30	Topical anesthesia (5% lidocaine)	VAS, heart rate	At least 2 weeks	Palatal: Cryotherapy = Topical anesthesia Buccal: Cryotherapy > Topical anesthesia
Hafeez, 2020	Split-mouth	9	F 4 M 5	20–60	Ice pack	1 min	NR	NR	Topical anesthesia spray (10% lidocaine)	VAS	NR	Cryotherapy > Topical anesthesia
Bansal, 2020	RCT	100	F 50 M 50	17–30	Ice finger	1 min	Anterior teeth infiltrations, maxilla	NR	Topical anesthesia (2% lidocaine)	VAS	NA	Cryotherapy > Topical anesthesia
Gadicherla, 2021	Split-mouth	70	F 41 M 29	24.77 (8.65)	Iced cotton buds	1 min	Greater palatine block and buccal infiltration/Maxilla	31	Topical anesthesia (2% lidocaine)	VAS	7–10 days	Cryotherapy = Topical anesthesia
Gurucharan, 2021	RCT	60	F 31 M 29	32.15 (8.06)	Ice stick	15 s	Buccal infiltration/Maxillary first molar	27	Topical anesthesia (2% lidocaine)	HP-VAS	NA	Cryotherapy > Topical anesthesia
AmruthaVarshini, 2021	RCT	30	NR	34.5 (9.0)	Ice finger	NR	Buccal infiltration/Primary maxillary posterior teeth	NR	Topical anesthesia (20% benzocaine), laser biostimulation	WB-FPRS, SEM	NA	Cryotherapy = Topical anesthesia
Kumari, 2021	Split-mouth	200	F 62 M 138	18–60	Ice cubes	2 min	Palatal nerve block/Maxilla	25	Topical anesthesia (lidocaine)	NRS, SEM	NR	Cryotherapy = Topical anesthesia

TABLE 3 (Continued)

Study ID	Study design	Sample size, no.	Gender, no.	Mean (SD) age, years	Cryotherapy type	Application time	Type/Site of anesthesia	Needle gauge	Comparison	Assessment type	Wash out period	Main results
Menon, 2021	Split-mouth	60	F 32 M 28	26.94 (0.76)	Refrigerant spray (tetrafluoroethene)	5 s	Inferior alveolar nerve block/ Mandible	30	Topical anesthesia (2% lidocaine)	VAS	NR	Cryotherapy > Topical anesthesia
Mohsin, 2022	Split-mouth	30	NR	18–55	Ice was prepared by filling a 2.5 mL plastic syringe halfway with mineral water and freezing it in the freezer	1 and 2.5 min	Buccal mucosa/ NR	30	Topical anesthesia (20% benzocaine)	VAS	2 weeks	Cryotherapy > Topical anesthesia
Abbasi, 2023_2	RCT	90	F 43 M 47	29.06 (5.7)	Cotton ball sprayed with ethyl chloride	30 s	Buccal infiltration/ Maxilla	27	Control, topical anesthesia (5% lidocaine)	VAS	NA	Cryotherapy = Topical anesthesia > Control
Abbasi, 2023_1	RCT	90	F 43 M 47	28.32 (6.39)	Cotton ball sprayed with ethyl chloride	30 s	Buccal infiltration/ Maxilla	27	Control, honey	VAS	NA	Cryotherapy > Honey > Control
Hemavathi, 2023	Split-mouth	20	F 7 M 13	46.9 (17.35)	Ice cubes	2 min	Palatal nerve block/Maxilla	25	EMLA 5% cream (lidocaine 2.5% w/w {25 mg/g} + Prilocaine 2.5% w/w {25 mg/g})	VAS, SEM	1 week	EMLA > Cryotherapy

Note: > superior effectiveness; = comparable effectiveness.
Abbreviations: F, female; HP-VAS, Heft Parker-Visual Analogue Scale; M, male; NA, not applicable; NR, not reported; NRS, Numerical Rating Scale; RCT, randomized clinical trial; SEM, Sound Eye Motor; VAS, Visual Analogue Scale; WB-FPRS, Wong Baker's Faces Pain Rating Scale.

Per-protocol	Unique ID	D1	D2	D3	D4	D5	Overall	
Abbasi 2023-1		+	+	+	!	+	!	+
Abbasi 2023-2		+	+	+	!	+	!	!
Aminabadi 2009		!	+	-	!	!	-	-
Aminah 2017		+	+	+	!	!	!	!
AmruthaVarshini 2021		+	+	+	!	+	!	!
Anantharaj 2020		!	+	+	!	!	!	!
Bansal 2020		!	+	+	!	!	!	!
Bhadoria 2017		!	+	+	!	!	!	!
Bose 2019		!	+	+	!	!	!	!
Chilakamuri 2020		+	+	+	!	!	!	!
Dhingra 2022		!	+	-	!	!	-	-
Gadicherla 2021		+	+	+	!	+	!	!
Ghaderi 2013		+	+	+	!	!	!	!
Gurucharan 2021		+	+	+	!	+	!	!
Hafeez 2020		!	+	+	!	!	!	!
Hameed 2018		-	+	-	!	!	-	-
Havale 2021		+	+	+	!	!	!	!
Hemavathi 2023		+	+	+	!	!	!	!
Hindocha 2019		+	+	+	!	+	!	!
Islam 2021		+	+	+	!	!	!	!
JafarTehrani 2021		!	+	+	!	+	!	!
Kosaraju 2009		+	+	+	!	!	!	!
Kumari 2021		!	+	+	!	!	!	!
Lakshmanan 2021		+	+	+	!	!	!	!
Lathwal 2015		+	+	-	!	!	-	-
Menon 2021		+	+	+	!	!	!	!
Mittal 2023		+	+	+	!	+	!	!
Mohiuddin 2015		-	+	+	!	!	-	-
Mohsin 2022		+	+	+	!	!	!	!
Nat 2023		!	+	+	!	!	!	!
Padmanabhan 2019		+	+	+	!	!	!	!
Prathyusha 2021		!	+	+	!	!	!	!
Singh 2022		+	+	+	!	!	!	!
Soni 2020		+	+	+	!	!	!	!
Vafaei 2019		!	+	-	!	+	-	-
Virdikar 2022		!	+	+	!	!	!	!
Wiswall 2014		+	+	+	!	!	!	!

Low risk
 Some concerns
 High risk

D1 Randomisation process
D2 Deviations from the intended interventions
D3 Missing outcome data
D4 Measurement of the outcome
D5 Selection of the reported result

FIGURE 2 Risk of bias summary using ROB-2.

children in the ice precooling group exhibited better behavioral responses compared with those in the refrigerant spray group. The ice was more familiar and tangible for the children than an applicator soaked in an unknown substance. As a result, children were more likely to accept the ice during the “tell, show, do” explanation process. This familiarity with ice could reduce anxiety and lead to improved behavioral feedback.

The experience of pain during dental infiltrations varies based on the oral cavity location. Distinct nerve pathways and tissue structures involved in anterior and posterior infiltrations may contribute to differing pain perceptions during injections. Nevertheless, due to a lack of evidence, a subgroup analysis for anterior and posterior infiltrations was deemed unfeasible. Notably, the study by Bansal et al.,²¹ focusing on anterior maxillary infiltrations,

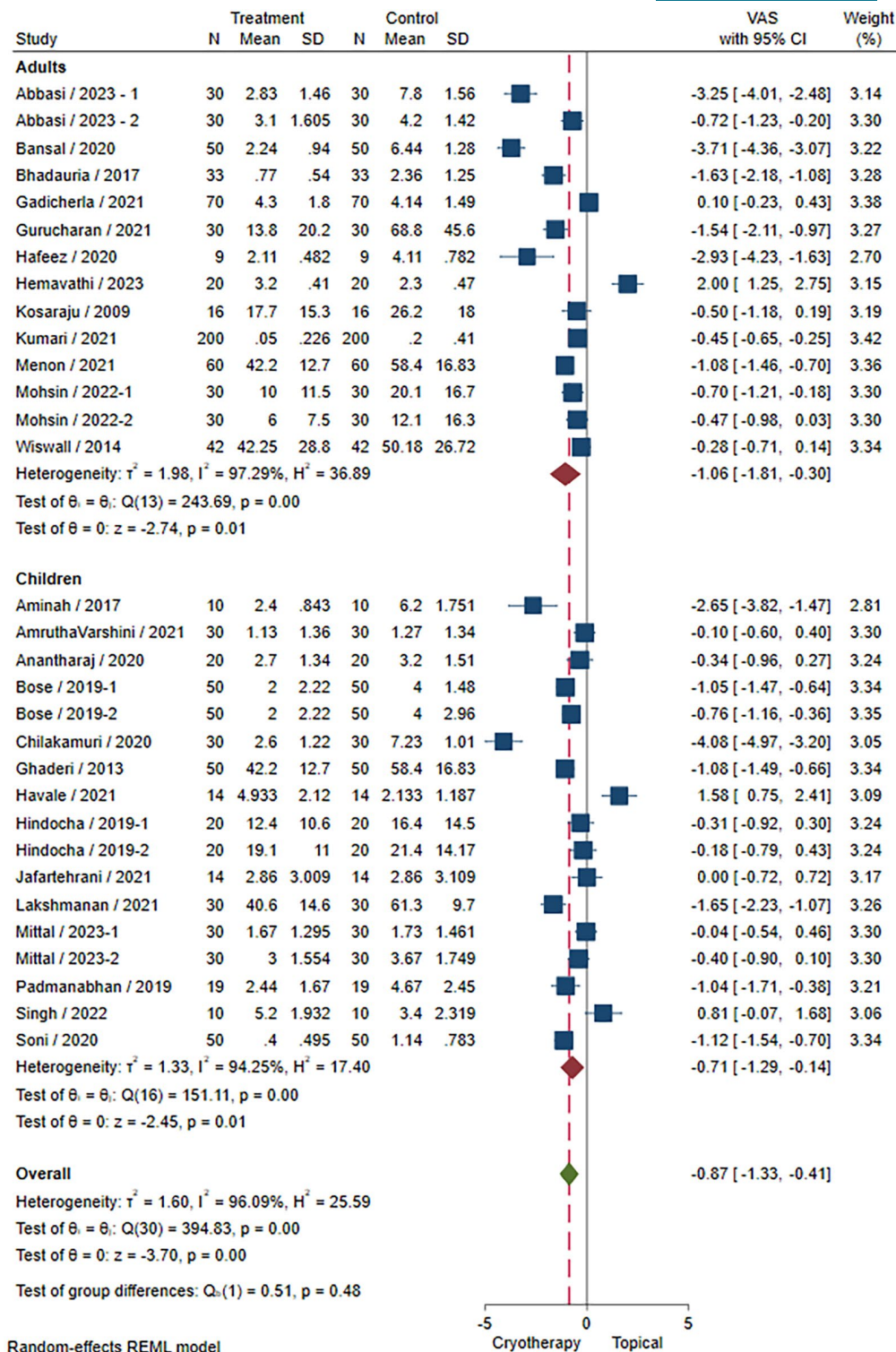


FIGURE 3 Forrest plot of random-effects analysis of standard mean differences between cryotherapy and topical anesthesia for Visual Analogue Scale (VAS).

recognized as the most painful buccal infiltration site, demonstrated superior pain reduction in cryotherapy application compared with topical anesthesia.

Furthermore, palatal injections are acknowledged as one of the most painful injection techniques due to the

proximity of palatal mucosa to the periosteum of the bone.⁵⁸ The greater palatine nerve block is also comparatively more painful than other nerve blocks.⁵⁹ Although some studies included palatal injections, only a few exclusively assessed them.^{10,12,26,30,35} Among these, three

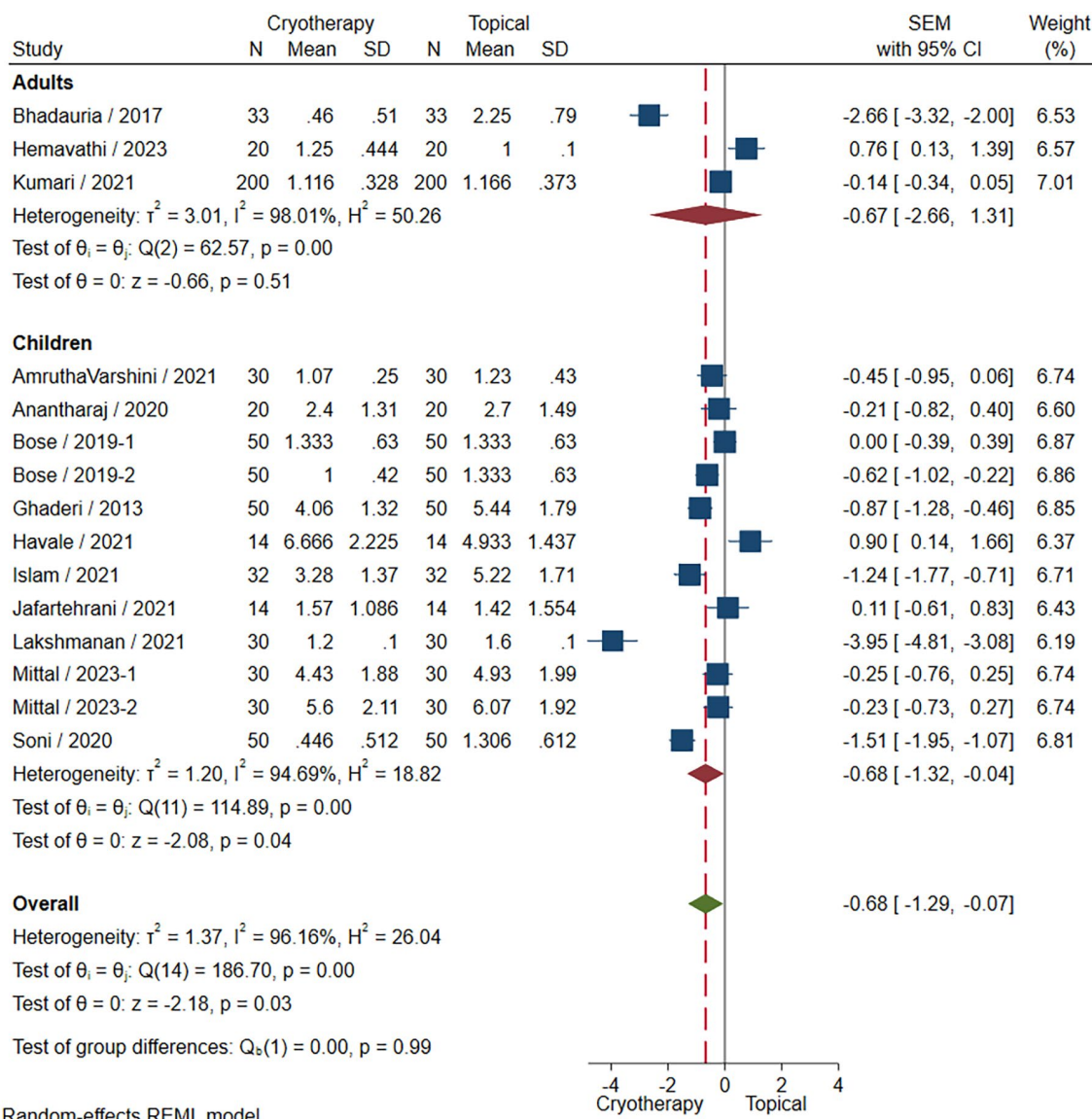


FIGURE 4 Forrest plot of random-effects analysis of standard mean differences between cryotherapy and topical anesthesia for Sound Eye Motor (SEM).

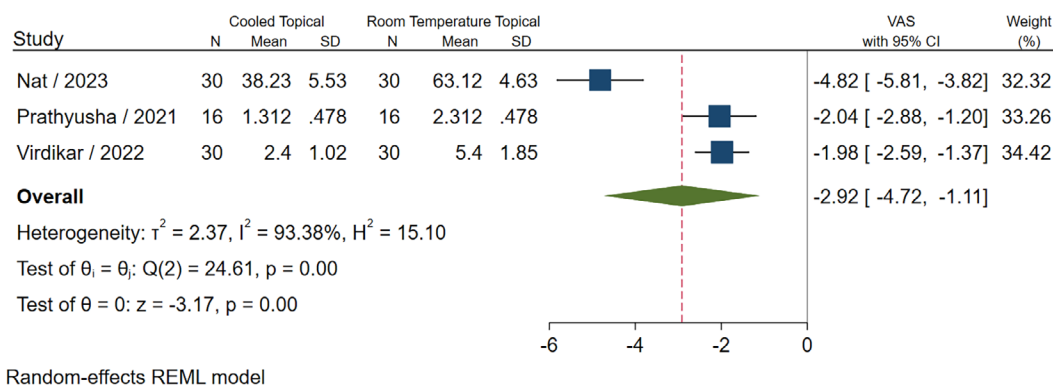


FIGURE 5 Forrest plot of random-effects analysis of standard mean differences between cooled topical anesthesia and room temperature topical anesthesia for Visual Analogue Scale (VAS).

studies by Kumari et al., Wiswall et al., and Gadicherla et al., assessing greater palatine nerve block in adults, found no significant difference between cryotherapy and topical anesthesia.^{10,25,35} The study by Chilakamuri et al.,¹² which assessed greater palatine nerve block in children, however, found superior pain reduction with cryotherapy application, compared with topical anesthesia.

Conversely, in another study by Hemavathi et al.,³⁰ evaluating palatal nerve blocks in adults, topical anesthesia showed greater superiority compared with cryotherapy. This discrepancy might be attributed to the uneven distribution of different types of palatal nerve blocks in the two groups, with the cryotherapy group including a significantly higher number of nasopalatine nerve blocks, recognized as the most painful palatal nerve block,⁶⁰ in contrast to the topical anesthesia group.

Despite the varying outcomes in the existing evidence and the necessity for additional studies to reach a firm conclusion, it seems that cryotherapy may not be as effective in palatal injections when compared to buccal injections.

Additionally, in the reviewed studies, a range of needle gauges spanning from 24 to 31 was utilized. Investigating the correlation between needle gauge and pain perception during intraoral injections is an ongoing area of research. Despite the common assumption that the use of smaller diameter (higher number gauge), needles might be less traumatic and lead to reduced pain, evidence supporting this notion appears to be lacking.⁶¹ Hamburg⁶² demonstrated as early as 1972 that patients could not differentiate between needles of different gauges (23-, 25-, 27-, and 30-gauge), a finding later corroborated by others.^{63,64} A study by Gadicherla et al.,²⁵ included in our review, however, reported significantly lower pain scores with 31-gauge compared with 26-gauge needles specifically on the buccal side.

Moreover, the literature reports instances of allergic responses to dental topical anesthetics,⁶⁵ underscoring the importance of considering alternative approaches to ensure patient safety. On the contrary, our review highlights cryotherapy as a safe and well-tolerated option for everyone and eliminates the risk of allergic reactions associated with topical anesthesia.

4.3 | Strengths, limitations, and recommendations for future research

Our review's strength lies in its comprehensive search for relevant studies and the inclusion of RCTs, bolstering the reliability of our findings. Moreover, the incorporation of 31 studies in the meta-analysis adds a considerable level of generalizability to our findings, enhancing the robustness and reliability of the results.

Our review, however, is not without limitations. Firstly, variations in study design, patient demographics, sample sizes, and protocols across the included studies might confound comparability, contributing to high heterogeneity and variability. Another limitation worth noting is the inherent subjectivity of pain perception, whether reported by patients (VAS) or observers (SEM). This subjectivity is influenced by individual factors such as age, gender, cultural background, and prior experiences, further complicating the generalizability of our findings.

Furthermore, the absence of sufficient evidence made a subgroup analysis based on gender and injection locations unfeasible, and future studies should address these aspects comprehensively. Moreover, all of the included studies exhibited a risk of bias, primarily due to the subjective nature of the measurements. Additional factors contributing to bias included flawed randomization processes and concerns regarding selective reporting. The certainty of the results, as assessed by the GRADE system, was determined to be of very low quality. This low certainty was largely due to high heterogeneity and the presence of biased studies, which should be considered when interpreting the findings. Additionally, the establishment of a standardized application time for cryotherapy in future research would contribute significantly to the advancement of this field.

In conclusion, our findings suggest that cryotherapy serves as an effective and potentially superior alternative to topical anesthesia, providing clinicians with an accessible and effective option for pain management during intraoral local anesthesia injections. The certainty of the results, however, is of very low quality. Pain perception is a complex and subjective experience, and generalizations may not apply to every individual. Factors such as gender, age, cultural background, and personal experiences can also play a significant role in shaping an individual's response to pain.

AUTHOR CONTRIBUTIONS

P.S. and M.G. conceived the ideas and designed the review. A.S.S. developed a search strategy and conducted the electronic search. P.S., M.G., and A.S.S. performed the screening. P.S. and M.G. performed data extraction. P.S. and A.S.S. performed a risk of bias assessment and statistical analysis. P.S. and M.G. prepared the initial draft. A.S.S. finalized the manuscript and critically reviewed the work/manuscript.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

The datasets used and/or analyzed during this study are available from the corresponding author upon reasonable request.

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